Additional file 1

Outcomes in all tables: PLE-HM, psychotic-like and hypomanic experiences at age 18; DEP, depressive symptoms at age 18 (both scaled from 0 to 1). GLMM, generalised linear mixed model; GAMM, generalised additive mixed model; RMSE, root-mean-squared error; SHAP, Shapley additive explanations.

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Table S1. DCMFNet hyperparameter search

Hyperparameters were searched with Optuna (tree-structured Parzen estimator sampler with default settings, no fixed sampler seed; median pruner with 5 start-up trials and 5 warm-up epochs), 50 trials per outcome. Each trial trained the network on 80% of the seed-42 training set (split by twin pair) and was scored by the lowest validation RMSE on the remaining 20%. In the search, one fusion depth applied to all blocks. Of the 50 trials, 28 (PLE-HM) and 32 (DEP) ran to completion and the rest were pruned. The best validation RMSE was 0.0880 (PLE-HM) and 0.1484 (DEP). The selected values were used for the final models, except that the PLE-HM fusion depths were then set per block (Table S5).

Hyperparameter	Search range	PLE-HM selected	DEP selected
Learning rate	10^{-5} to 10^{-2} (log scale)	1.00e-04	3.22e-04
Batch size	4, 6, 8, 16, 32	4	6
Maximum epochs	10 to 30	23	20
Weight decay	10^{-5} to 10^{-2} (log scale)	2.95e-05	1.79e-04
Fusion depth (all blocks)	1 to 7	3 (replaced by per-block depths, Table S5)	2
Dropout	0.1 to 0.5	0.234	0.326
Gate reduction ratio r	2 to 8	6	5
Gate minimum width h_min	2 to 16	13	5
Base loss	MSE, Huber	Huber	Huber
Huber δ	0.01 to 0.2	0.086	0.116
Focal γ	0.5 to 3.0	1.32	2.62
Bins K	5 to 20	13	10
Scheduler patience	2 to 5	4	2
Scheduler factor	0.3 to 0.7	0.432	0.549
Early-stopping patience	3 to 8	5	7

Tuning history. The search above is one of at least five tuning runs carried out over five days, which differed in their search spaces (for example, uniform or per-block depths). The hyperparameters used in this study come from the run shown here; the choice among runs also took the test-set performance of the resulting models into account. The outputs of the other runs are kept in the version history of the code repository. Because of this, and because the seed-42 training set used for tuning overlaps the test sets of the other four splits, DCMFNet's test performance may be optimistic (see Methods and the limitations in the Discussion).

Table S2. Default-settings screening of 40 machine-learning algorithms

Test-set R^2 and Spearman's ρ, mean (SD) over the five twin-pair-grouped splits used for all models. Each algorithm was fitted with its scikit-learn default settings on the 105 unscaled inputs, without tuning and without modelling twin pairs; estimators with a built-in penalty search (for example LassoCV, RidgeCV, LassoLarsIC) chose their penalty within the training split, without grouping by twin pair. Algorithms are ordered by R^2 for PLE-HM. These results describe default settings only and were not used to select models.

Algorithm	PLE-HM R^2	PLE-HM Spearman ρ	DEP R^2	DEP Spearman ρ
LassoCV	0.261 (0.019)	0.492 (0.019)	0.234 (0.017)	0.472 (0.020)
LassoLarsCV	0.261 (0.019)	0.492 (0.019)	0.234 (0.017)	0.472 (0.020)
ElasticNetCV	0.261 (0.019)	0.492 (0.019)	0.234 (0.017)	0.471 (0.020)
LassoLarsIC	0.261 (0.019)	0.492 (0.019)	0.234 (0.017)	0.471 (0.019)
LarsCV	0.259 (0.021)	0.491 (0.018)	0.235 (0.016)	0.472 (0.019)
BayesianRidge	0.255 (0.021)	0.486 (0.022)	0.226 (0.020)	0.461 (0.020)
RidgeCV	0.246 (0.022)	0.478 (0.022)	0.218 (0.024)	0.456 (0.021)
KernelRidge	0.244 (0.021)	0.475 (0.021)	0.215 (0.025)	0.454 (0.020)
Ridge	0.242 (0.021)	0.474 (0.021)	0.215 (0.025)	0.454 (0.021)
HuberRegressor	0.240 (0.022)	0.479 (0.020)	0.206 (0.023)	0.454 (0.022)
TransformedTargetRegressor	0.239 (0.021)	0.473 (0.022)	0.215 (0.025)	0.453 (0.020)
LinearRegression	0.239 (0.021)	0.473 (0.022)	0.215 (0.025)	0.453 (0.020)
Lars	0.236 (0.022)	0.470 (0.021)	0.214 (0.025)	0.452 (0.019)
GradientBoostingRegressor	0.229 (0.019)	0.461 (0.015)	0.214 (0.012)	0.455 (0.019)
SGDRegressor	0.228 (0.029)	0.469 (0.021)	0.208 (0.018)	0.451 (0.017)
OrthogonalMatchingPursuitCV	0.225 (0.023)	0.463 (0.018)	0.227 (0.015)	0.462 (0.019)
OrthogonalMatchingPursuit	0.225 (0.023)	0.463 (0.018)	0.227 (0.015)	0.462 (0.019)
RandomForestRegressor	0.221 (0.021)	0.455 (0.013)	0.185 (0.016)	0.421 (0.027)
TweedieRegressor	0.211 (0.016)	0.458 (0.021)	0.181 (0.013)	0.438 (0.018)
ExtraTreesRegressor	0.209 (0.026)	0.446 (0.018)	0.138 (0.030)	0.396 (0.035)
HistGradientBoostingRegressor	0.199 (0.015)	0.438 (0.006)	0.174 (0.025)	0.420 (0.026)
LGBMRegressor	0.196 (0.018)	0.433 (0.010)	0.182 (0.029)	0.426 (0.030)
SVR	0.176 (0.015)	0.408 (0.008)	0.155 (0.026)	0.413 (0.025)
NuSVR	0.158 (0.027)	0.423 (0.014)	0.140 (0.026)	0.412 (0.024)
BaggingRegressor	0.132 (0.029)	0.374 (0.021)	0.116 (0.019)	0.380 (0.014)
PoissonRegressor	0.110 (0.008)	0.435 (0.021)	0.114 (0.010)	0.413 (0.021)
LinearSVR	0.070 (0.077)	0.439 (0.036)	0.101 (0.056)	0.423 (0.018)
KNeighborsRegressor	0.034 (0.014)	0.315 (0.017)	0.018 (0.031)	0.302 (0.030)
AdaBoostRegressor	0.028 (0.034)	0.387 (0.029)	0.005 (0.033)	0.420 (0.015)
PassiveAggressiveRegressor‡	0.024 (0.249)	0.404 (0.034)	−0.226 (0.511)	0.405 (0.035)
QuantileRegressor†	−0.000 (0.000)	—	−0.052 (0.056)	—
ElasticNet†	−0.001 (0.001)	—	−0.002 (0.002)	—
LassoLars†	−0.001 (0.001)	—	−0.002 (0.002)	—
DummyRegressor†	−0.001 (0.001)	—	−0.002 (0.002)	—
Lasso†	−0.001 (0.001)	—	−0.002 (0.002)	—
MLPRegressor‡	−0.359 (0.144)	0.295 (0.022)	−0.344 (0.153)	0.289 (0.020)
DecisionTreeRegressor‡	−0.574 (0.066)	0.210 (0.018)	−0.552 (0.073)	0.210 (0.007)
ExtraTreeRegressor‡	−0.578 (0.111)	0.226 (0.030)	−0.575 (0.067)	0.224 (0.020)
GaussianProcessRegressor‡	−1.567 (0.043)	−0.329 (0.008)	−1.759 (0.093)	−0.213 (0.032)
RANSACRegressor‡	−9.902 (4.375)	0.185 (0.034)	−6.420 (1.827)	0.126 (0.030)

† Predicted a constant (or nearly constant) value on at least one outcome, so Spearman's ρ is undefined (—); for the lasso, elastic net and LARS lasso this is the effect of the default penalty ($\alpha = 1$) on these outcome scales. ‡ Negative mean R^2 on at least one outcome: predictions were worse than the test-set mean.

Table S2b. Hyperparameters selected for the tuned comparators

Selected for each split, predictor set and outcome by 5-fold cross-validation within the training split, with folds grouped by twin pair and mean squared error (MSE) as the criterion. Linear regression has no hyperparameters. "All" uses all 105 predictors; "Age 15" omits adverse experiences and substance use at age 18.

Elastic net (inputs standardised; α is the penalty strength, L1 ratio 1 = lasso, close to 0 = ridge).

Predictors	Outcome	Split	α	L1 ratio	Inner-CV MSE ($\times 10^3$)	Test R^2
All	PLE-HM	42	0.0393	0.05	7.78	0.293
All	PLE-HM	43	0.0399	0.05	7.47	0.250
All	PLE-HM	44	0.0381	0.05	7.49	0.257
All	PLE-HM	45	0.0319	0.05	7.55	0.244
All	PLE-HM	46	0.0399	0.05	7.66	0.258
All	DEP	42	0.0086	0.30	22.74	0.258
All	DEP	43	0.0025	1.00	21.89	0.251
All	DEP	44	0.0275	0.10	22.49	0.230
All	DEP	45	0.0440	0.05	22.42	0.221
All	DEP	46	0.0039	1.00	22.63	0.226
Age 15	PLE-HM	42	0.0354	0.05	8.29	0.214
Age 15	PLE-HM	43	0.0419	0.05	8.02	0.184
Age 15	PLE-HM	44	0.0399	0.05	8.10	0.204
Age 15	PLE-HM	45	0.0300	0.05	8.19	0.192
Age 15	PLE-HM	46	0.0367	0.05	8.24	0.197
Age 15	DEP	42	0.0484	0.05	24.24	0.182
Age 15	DEP	43	0.0542	0.05	23.51	0.180
Age 15	DEP	44	0.0113	0.30	24.33	0.179
Age 15	DEP	45	0.0582	0.05	24.31	0.175
Age 15	DEP	46	0.0586	0.05	24.29	0.157

Gradient boosting (scikit-learn HistGradientBoostingRegressor; 40 random configurations per fit; early stopping off).

Predictors	Outcome	Split	Learning rate	Trees	Max depth	Max leaves	Min leaf size	L2 penalty	Inner-CV MSE ($\times 10^3$)	Test R^2
All	PLE-HM	42	0.032	225	2	5	72	0.00136	7.90	0.277
All	PLE-HM	43	0.026	352	2	10	115	0.00113	7.59	0.227
All	PLE-HM	44	0.014	487	6	4	104	2.18	7.71	0.254
All	PLE-HM	45	0.040	348	2	20	79	0.188	7.67	0.232
All	PLE-HM	46	0.031	256	4	6	62	0.456	7.79	0.231
All	DEP	42	0.032	225	2	5	72	0.00136	23.14	0.254
All	DEP	43	0.016	312	3	25	44	0.0158	22.27	0.239
All	DEP	44	0.014	487	6	4	104	2.18	22.87	0.225
All	DEP	45	0.051	292	2	12	109	0.257	22.79	0.207
All	DEP	46	0.031	256	4	6	62	0.456	23.07	0.219

Predictors	Outcome	Split	Learning rate	Trees	Max depth	Max leaves	Min leaf size	L2 penalty	Inner-CV MSE ($\times 10^3$)	Test R^2
Age 15	PLE-HM	42	0.043	158	2	13	34	0.126	8.38	0.205
Age 15	PLE-HM	43	0.029	288	2	30	132	0.00151	8.17	0.164
Age 15	PLE-HM	44	0.014	487	6	4	104	2.18	8.27	0.189
Age 15	PLE-HM	45	0.014	203	3	24	36	0.548	8.34	0.182
Age 15	PLE-HM	46	0.013	449	2	19	143	0.127	8.32	0.176
Age 15	DEP	42	0.019	317	3	15	181	0.154	24.68	0.169
Age 15	DEP	43	0.016	312	3	25	44	0.0158	23.58	0.161
Age 15	DEP	44	0.014	487	6	4	104	2.18	24.59	0.173
Age 15	DEP	45	0.051	292	2	12	109	0.257	24.64	0.156
Age 15	DEP	46	0.013	449	2	19	143	0.127	24.69	0.170

Table S3. Fixed effects and smooth terms of the mixed models

Estimates from the models fitted on the training set of the first split (seed 42). Outcomes are scaled to 0–1. Main effects entered on their original scales. The interaction terms use standardised scores: each group "score" is the mean of the group's items after standardising each item, standardised again, and each PRS in an interaction is standardised (z). Means and SDs came from the training set. p values are unadjusted. In the generalised additive mixed model (GAMM), each PRS entered as a thin-plate regression spline (Table S3c) instead of a linear term, and the twin-pair random intercept was a random-effect smooth. GLMM, generalised linear mixed model (Gaussian outcome, identity link).

Table S3a. PLE-HM: fixed effects of the linear and additive mixed models

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
Substance use, age 15 (anchor)				
SUD15_OtherDrugs15	0.0171 (0.0082)	0.0373	0.0175 (0.0082)	0.0333
SUD15_Painkillers_opioids15	0.0081 (0.0058)	0.163	0.0083 (0.0058)	0.153
SUD15_Snuff15	0.0055 (0.0049)	0.258	0.0055 (0.0049)	0.260
SUD15_Cannabis15	0.0102 (0.0092)	0.267	0.0108 (0.0092)	0.239
SUD15_Alcohol15	-0.0022 (0.0029)	0.460	-0.0018 (0.0029)	0.533
SUD15_Cigarettes15	-0.0011 (0.0043)	0.803	-0.0008 (0.0043)	0.853
Polygenic risk scores				
PRS_all	0.0199 (0.0056)	< 0.001	smooth, Table S3c	–
PRS_Miscellaneous	0.0366 (0.0189)	0.0531	smooth, Table S3c	–
PRS_Hippocampal_CA1_3	-0.0299 (0.0197)	0.129	smooth, Table S3c	–
PRS_Amygdala_excitatory	-0.0239 (0.0191)	0.211	smooth, Table S3c	–
PRS_Deep_layer_intratelencepha	-0.0231 (0.0205)	0.260	smooth, Table S3c	–
PRS_CGE_interneuron	-0.0165 (0.0186)	0.374	smooth, Table S3c	–
PRS_Cerebellar_inhibitory	-0.0143 (0.0176)	0.415	smooth, Table S3c	–
PRS_MGE_interneuron	-0.0116 (0.0187)	0.535	smooth, Table S3c	–
PRS_Deep_layer_corticothalamic	0.0098 (0.0185)	0.597	smooth, Table S3c	–
PRS_Hippocampal_CA4	0.0093 (0.0191)	0.626	smooth, Table S3c	–
PRS_Upper_layer_intratelenceph	0.0077 (0.0192)	0.688	smooth, Table S3c	–
PRS_Thalamic_excitatory	0.0055 (0.0168)	0.746	smooth, Table S3c	–
PRS_LAMP5_LHX6_and_Chandelier	-0.0058 (0.0192)	0.764	smooth, Table S3c	–
PRS_Medium_spiny_neuron	0.0047 (0.0192)	0.807	smooth, Table S3c	–

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
PRS_Hippocampal_dentate_gyrus	0.0033 (0.0181)	0.857	smooth, Table S3c	–
PRS_Eccentric_medium_spiny_neu	0.0004 (0.0179)	0.983	smooth, Table S3c	–
Symptoms, age 15				
SCZ15_Too_much_energy15	0.0108 (0.0024)	< 0.001	0.0109 (0.0024)	< 0.001
SCZ15_Seen_hallucinations15	0.0155 (0.0040)	< 0.001	0.0156 (0.0040)	< 0.001
SCZ15_Talking_fast15	0.0091 (0.0025)	< 0.001	0.0091 (0.0025)	< 0.001
SCZ15_Read_others_minds15	0.0083 (0.0024)	< 0.001	0.0083 (0.0024)	< 0.001
SCZ15_Extreme_excitement15	0.0059 (0.0019)	0.00175	0.0058 (0.0019)	0.00198
SCZ15_Hear_voices15	0.0206 (0.0067)	0.00201	0.0202 (0.0066)	0.00242
SCZ15_headaches15	0.0079 (0.0027)	0.00334	0.0079 (0.0027)	0.00363
SCZ15_Special_messages15	0.0187 (0.0064)	0.00354	0.0185 (0.0064)	0.00384
SCZ15_Spied15	0.0051 (0.0026)	0.0506	0.0052 (0.0026)	0.0493
SCZ15_Irritable15	0.0044 (0.0027)	0.0955	0.0044 (0.0027)	0.0973
SCZ15_Special_powers15	0.0084 (0.0058)	0.147	0.0081 (0.0058)	0.160
SCZ15_PROD_seen_hallucinations9	0.0117 (0.0094)	0.214	0.0122 (0.0094)	0.194
SCZ15_Rage_attacks15	0.0048 (0.0039)	0.220	0.0048 (0.0039)	0.218
SCZ15_Sexual_inappropriate15	0.0114 (0.0107)	0.283	0.0108 (0.0106)	0.310
SCZ15_Unrealistic_abilities15	0.0071 (0.0073)	0.330	0.0079 (0.0073)	0.280
SCZ15_Racing_thoughts15	0.0022 (0.0027)	0.425	0.0023 (0.0027)	0.405
SCZ15_easily_scared15	-0.0022 (0.0028)	0.444	-0.0019 (0.0028)	0.510
SCZ15_unhappy15	-0.0026 (0.0036)	0.464	-0.0028 (0.0036)	0.438
SCZ15_Others_Read_thoughts15	0.0017 (0.0028)	0.542	0.0016 (0.0028)	0.577
SCZ15_worry15	-0.0014 (0.0027)	0.603	-0.0015 (0.0027)	0.576
SCZ15_Under_control_special_powers15	0.0030 (0.0042)	0.645	0.0021 (0.0042)	0.624
SCZ15_Not_tired15	-0.0008 (0.0022)	0.715	-0.0009 (0.0022)	0.685
SCZ15_lose_confidence15	0.0009 (0.0025)	0.719	0.0010 (0.0025)	0.687
ADHD symptoms, age 9				
ADHD9_talk_excess9	0.0067 (0.0036)	0.0599	0.0068 (0.0036)	0.0547
ADHD9_easily_bored9	-0.0070 (0.0041)	0.0867	-0.0070 (0.0041)	0.0863
ADHD9_difficulty_organizing_tasks9	-0.0114 (0.0070)	0.102	-0.0116 (0.0070)	0.0956
ADHD9_not_listening9	-0.0058 (0.0049)	0.238	-0.0055 (0.0049)	0.260
ADHD9_often_motor9	0.0066 (0.0072)	0.355	0.0067 (0.0072)	0.348
ADHD9_easily_distracted9	0.0041 (0.0044)	0.359	0.0042 (0.0044)	0.339
ADHD9_avoid_mental_effort_tasks9	0.0037 (0.0049)	0.450	0.0035 (0.0049)	0.477
ADHD9_difficulty_with_attention9	0.0041 (0.0054)	0.456	0.0040 (0.0054)	0.461
ADHD9_difficulty_waiting9	0.0035 (0.0047)	0.459	0.0033 (0.0047)	0.480
ADHD9_often_run9	-0.0034 (0.0053)	0.513	-0.0033 (0.0053)	0.529
ADHD9_often_interrupt9	-0.0033 (0.0053)	0.539	-0.0029 (0.0053)	0.581
ADHD9_often_lose_things9	0.0020 (0.0043)	0.638	0.0018 (0.0043)	0.676
ADHD9_difficulty_following_instructions9	-0.0023 (0.0070)	0.739	-0.0024 (0.0070)	0.734
ADHD9_often_forgetful9	0.0016 (0.0050)	0.752	0.0016 (0.0050)	0.748
ADHD9_often_move9	0.0013 (0.0060)	0.830	0.0010 (0.0060)	0.862
ADHD9_often_blurt_out9	-0.0009 (0.0045)	0.851	-0.0014 (0.0045)	0.764
ADHD9_difficulty_hold_still9	0.0007 (0.0049)	0.884	0.0008 (0.0049)	0.875
ADHD9_fail_close_attention9	0.0006 (0.0049)	0.898	0.0006 (0.0049)	0.909
ADHD9_difficulty_calm9	-0.0009 (0.0086)	0.918	-0.0004 (0.0086)	0.963

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
Autism-related traits, age 9				
ASD9_difficulty_converse9	-0.0324 (0.0124)	0.00929	-0.0323 (0.0124)	0.00936
ASD9_easily_influenced9	0.0086 (0.0038)	0.0222	0.0085 (0.0038)	0.0242
ASD9_difficulty_socialize9	0.0224 (0.0099)	0.0241	0.0212 (0.0099)	0.0326
ASD9_absorbed_details9	0.0134 (0.0060)	0.0243	0.0134 (0.0059)	0.0239
ASD9_absorbed_problems9	-0.0159 (0.0098)	0.106	-0.0157 (0.0098)	0.109
ASD9_difficulty_pretend_play9	-0.0154 (0.0096)	0.110	-0.0152 (0.0096)	0.114
ASD9_delay_language9	0.0067 (0.0043)	0.120	0.0067 (0.0043)	0.121
ASD9_uninterested_sharing9	0.0165 (0.0130)	0.206	0.0161 (0.0130)	0.214
ASD9_difficulty_expect_behavior9	-0.0120 (0.0096)	0.215	-0.0116 (0.0096)	0.230
ASD9_absorbed_own9	-0.0046 (0.0059)	0.436	-0.0044 (0.0059)	0.461
ASD9_difficulty_on_track9	0.0045 (0.0062)	0.469	0.0046 (0.0062)	0.461
ASD9_talk_too_high_low9	-0.0028 (0.0051)	0.581	-0.0029 (0.0051)	0.573
ASD9_own_terms9	0.0028 (0.0071)	0.697	0.0028 (0.0071)	0.694
ASD9_dislike_change9	0.0017 (0.0050)	0.736	0.0014 (0.0050)	0.785
ASD9_strange_movements9	-0.0020 (0.0069)	0.774	-0.0022 (0.0069)	0.747
ASD9_difficulty_express9	0.0040 (0.0142)	0.778	0.0041 (0.0142)	0.775
ASD9_repeat_words9	-0.0008 (0.0074)	0.911	-0.0004 (0.0074)	0.954
Adverse experiences, age 15				
ACE15_bullying_by_num15	-0.0065 (0.0034)	0.0554	-0.0065 (0.0034)	0.0526
ACE15_rumours_bullying15	0.0078 (0.0043)	0.0695	0.0078 (0.0043)	0.0691
ACE15_other_bullying15	0.0054 (0.0038)	0.153	0.0053 (0.0038)	0.161
ACE15_emotional_bullying15	0.0035 (0.0036)	0.321	0.0033 (0.0036)	0.353
ACE15_tease_bullying15	0.0015 (0.0040)	0.706	0.0014 (0.0040)	0.727
ACE15_bullying_time15	0.0008 (0.0023)	0.734	0.0010 (0.0023)	0.678
ACE15_bullied_often15	-0.0001 (0.0041)	0.988	0.0001 (0.0041)	0.971
Adverse experiences, age 18				
ACE18_emotional_abuse18	-0.0353 (0.0039)	< 0.001	-0.0355 (0.0039)	< 0.001
ACE18_witness_crime18	-0.0216 (0.0060)	< 0.001	-0.0215 (0.0060)	< 0.001
ACE18_hate_crime18	-0.0197 (0.0103)	0.0549	-0.0198 (0.0103)	0.0542
ACE18_other_abuse18	-0.0476 (0.2136)	0.824	-0.0409 (0.2134)	0.848
Substance use, age 18				
SUD18_alcohol_often18	0.0094 (0.0024)	< 0.001	0.0093 (0.0024)	< 0.001
SUD18_drugs_often18	0.0165 (0.0049)	< 0.001	0.0163 (0.0048)	< 0.001
SUD18_cigarettes18	0.0037 (0.0012)	0.00251	0.0037 (0.0012)	0.00221
SUD18_snuff18	0.0008 (0.0010)	0.419	0.0008 (0.0010)	0.401
Parental socioeconomic indicators				
SES_education_father	-0.0026 (0.0017)	0.117	-0.0027 (0.0017)	0.109
SES_birth_country_father	-0.0035 (0.0041)	0.398	-0.0036 (0.0041)	0.375
SES_birth_country_mother	-0.0009 (0.0046)	0.843	-0.0009 (0.0046)	0.851
SES_education_mother	0.0001 (0.0019)	0.946	0.0002 (0.0019)	0.926
Sex				
SEX	-0.0037 (0.0040)	0.360	-0.0037 (0.0040)	0.355
Genotyping batch $\times$ principal component				

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
batch_GSA3_x_PC1	-0.3202 (0.2236)	0.152	-0.3106 (0.2232)	0.164
batch_PSYC_x_PC2	-0.1883 (0.1322)	0.154	-0.1796 (0.1320)	0.174
batch_PSYC_x_PC1	-0.0755 (0.1387)	0.586	-0.0635 (0.1385)	0.647
batch_GSA3_x_PC2	0.0351 (0.3762)	0.926	0.0352 (0.3753)	0.925
Interactions with SUD15				
SUD15 score × ACE15 score	-0.0021 (0.0018)	0.231	-0.0021 (0.0018)	0.232
SUD15 score × ACE18 score	-0.0001 (0.0016)	0.974	-0.0001 (0.0016)	0.941
SUD15 score × ADHD9 score	0.0022 (0.0016)	0.178	0.0022 (0.0016)	0.186
SUD15 score × ASD9 score	-0.0021 (0.0018)	0.251	-0.0021 (0.0018)	0.265
SUD15 score × SCZ15 score	0.0019 (0.0016)	0.226	0.0020 (0.0016)	0.201
SUD15 score × SES score	-0.0014 (0.0019)	0.450	-0.0014 (0.0019)	0.466
SUD15 score × SEX	-0.0039 (0.0040)	0.330	-0.0042 (0.0040)	0.293
SUD15 score × SUD18 score	-0.0006 (0.0017)	0.722	-0.0007 (0.0017)	0.703
SUD15 score × PRS_Amygdala_excitatory (z)	0.0038 (0.0027)	0.151	0.0036 (0.0027)	0.173
SUD15 score × PRS_CGE_interneuron (z)	-0.0014 (0.0024)	0.579	-0.0015 (0.0024)	0.542
SUD15 score × PRS_Cerebellar_inhibitory (z)	-0.0013 (0.0022)	0.549	-0.0013 (0.0022)	0.563
SUD15 score × PRS_Deep_layer_corticothalamic (z)	0.0011 (0.0024)	0.630	0.0011 (0.0024)	0.631
SUD15 score × PRS_Deep_layer_intratelencepha (z)	0.0006 (0.0028)	0.839	0.0007 (0.0028)	0.801
SUD15 score × PRS_Eccentric_medium_spiny_neu (z)	-0.0019 (0.0025)	0.442	-0.0019 (0.0025)	0.436
SUD15 score × PRS_Hippocampal_CA1_3 (z)	-0.0011 (0.0028)	0.710	-0.0012 (0.0028)	0.674
SUD15 score × PRS_Hippocampal_CA4 (z)	-0.0048 (0.0025)	0.055	-0.0048 (0.0025)	0.0554
SUD15 score × PRS_Hippocampal_dentate_gyrus (z)	0.0018 (0.0022)	0.436	0.0018 (0.0022)	0.426
SUD15 score × PRS_LAMP5_LHX6_and_Chandelier (z)	0.0002 (0.0028)	0.944	0.0004 (0.0028)	0.888
SUD15 score × PRS_MGE_interneuron (z)	0.0053 (0.0025)	0.0348	0.0053 (0.0025)	0.0341
SUD15 score × PRS_Medium_spiny_neuron (z)	0.0061 (0.0026)	0.0184	0.0061 (0.0026)	0.0183
SUD15 score × PRS_Miscellaneous (z)	-0.0009 (0.0025)	0.723	-0.0007 (0.0025)	0.776
SUD15 score × PRS_Thalamic_excitatory (z)	-0.0007 (0.0019)	0.730	-0.0007 (0.0019)	0.692

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
SUD15 score × PRS_Upper_layer_intratelenceph (z)	-0.0054 (0.0025)	0.0301	-0.0052 (0.0025)	0.0378
SUD15 score × PRS_all (z)	0.0009 (0.0022)	0.677	0.0010 (0.0022)	0.664

Table S3b. DEP: fixed effects of the linear and additive mixed models

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
Substance use, age 15 (anchor)				
SUD15_Cigarettes15	-0.0152 (0.0073)	0.0368	-0.0151 (0.0073)	0.0389
SUD15_Snuff15	0.0065 (0.0083)	0.438	0.0063 (0.0083)	0.450
SUD15_Alcohol15	0.0021 (0.0050)	0.668	0.0022 (0.0050)	0.653
SUD15_OtherDrugs15	-0.0060 (0.0140)	0.671	-0.0059 (0.0140)	0.675
SUD15_Painkillers_opioids15	-0.0010 (0.0099)	0.923	-0.0008 (0.0099)	0.933
SUD15_Cannabis15	-0.0009 (0.0157)	0.953	-0.0006 (0.0157)	0.972
Polygenic risk scores				
PRS_all	0.0238 (0.0097)	0.0142	smooth, Table S3c	–
PRS_Hippocampal_CA4	-0.0415 (0.0329)	0.208	smooth, Table S3c	–
PRS_Cerebellar_inhibitory	0.0321 (0.0303)	0.289	smooth, Table S3c	–
PRS_Eccentric_medium_spiny_neu	0.0310 (0.0309)	0.315	smooth, Table S3c	–
PRS_Medium_spiny_neuron	-0.0298 (0.0331)	0.367	smooth, Table S3c	–
PRS_Deep_layer_intratelencepha	-0.0288 (0.0352)	0.414	smooth, Table S3c	–
PRS_Deep_layer_corticothalamic	0.0242 (0.0318)	0.447	smooth, Table S3c	–
PRS_MGE_interneuron	-0.0191 (0.0323)	0.554	smooth, Table S3c	–
PRS_Amygdala_excitatory	0.0172 (0.0329)	0.602	smooth, Table S3c	–
PRS_LAMP5_LHX6_and_Chandelier	-0.0144 (0.0331)	0.664	smooth, Table S3c	–
PRS_Miscellaneous	0.0080 (0.0326)	0.807	smooth, Table S3c	–
PRS_Hippocampal_dentate_gyrus	0.0072 (0.0312)	0.817	smooth, Table S3c	–
PRS_Thalamic_excitatory	0.0040 (0.0290)	0.889	smooth, Table S3c	–
PRS_Upper_layer_intratelenceph	0.0030 (0.0330)	0.927	smooth, Table S3c	–
PRS_Hippocampal_CA1_3	0.0019 (0.0339)	0.955	smooth, Table S3c	–
PRS_CGE_interneuron	0.0001 (0.0320)	0.997	smooth, Table S3c	–
Symptoms, age 15				
SCZ15_unhappy15	0.0418 (0.0061)	< 0.001	0.0415 (0.0061)	< 0.001
SCZ15_headaches15	0.0286 (0.0046)	< 0.001	0.0283 (0.0046)	< 0.001
SCZ15_worry15	0.0218 (0.0047)	< 0.001	0.0219 (0.0047)	< 0.001
SCZ15_Extreme_excitement15	-0.0111 (0.0032)	< 0.001	-0.0114 (0.0032)	< 0.001
SCZ15_Irritable15	0.0145 (0.0045)	0.00142	0.0145 (0.0045)	0.00141
SCZ15_lose_confidence15	0.0112 (0.0044)	0.0103	0.0113 (0.0044)	0.00977
SCZ15_Too_much_energy15	0.0105 (0.0041)	0.0107	0.0103 (0.0041)	0.0124
SCZ15_Hear_voices15	0.0182 (0.0114)	0.110	0.0177 (0.0114)	0.120
SCZ15_Unrealistic_abilities15	0.0171 (0.0124)	0.168	0.0171 (0.0124)	0.167
SCZ15_Special_messages15	0.0149 (0.0109)	0.172	0.0147 (0.0109)	0.179
SCZ15_Rage_attacks15	0.0085 (0.0066)	0.201	0.0086 (0.0066)	0.193

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
SCZ15_Seen_hallucinations15	0.0080 (0.0069)	0.245	0.0081 (0.0069)	0.235
SCZ15_Read_others_minds15	-0.0042 (0.0042)	0.319	-0.0042 (0.0042)	0.310
SCZ15_Others_Read_thoughts15	0.0047 (0.0048)	0.325	0.0044 (0.0048)	0.361
SCZ15_Racing_thoughts15	0.0039 (0.0047)	0.407	0.0041 (0.0047)	0.381
SCZ15_PROD_seen_hallucinations9	-0.0094 (0.0161)	0.557	-0.0090 (0.0161)	0.576
SCZ15_Spied15	-0.0026 (0.0045)	0.567	-0.0024 (0.0045)	0.599
SCZ15_Special_powers15	-0.0052 (0.0099)	0.599	-0.0053 (0.0099)	0.593
SCZ15_Under_control_special_powers15	-0.0033 (0.0072)	0.644	-0.0033 (0.0072)	0.644
SCZ15_Sexual_inappropriate15	0.0074 (0.0182)	0.686	0.0078 (0.0182)	0.669
SCZ15_Talking_fast15	-0.0013 (0.0043)	0.771	-0.0012 (0.0043)	0.787
SCZ15_easily_scared15	0.0014 (0.0048)	0.773	0.0014 (0.0048)	0.776
SCZ15_Not_tired15	0.0002 (0.0037)	0.956	0.0001 (0.0037)	0.985
ADHD symptoms, age 9				
ADHD9_often_blurt_out9	-0.0183 (0.0077)	0.0184	-0.0182 (0.0077)	0.0187
ADHD9_fail_close_attention9	-0.0160 (0.0083)	0.0542	-0.0159 (0.0083)	0.0554
ADHD9_often_interrupt9	0.0148 (0.0091)	0.103	0.0146 (0.0091)	0.107
ADHD9_often_forgetful9	0.0126 (0.0086)	0.141	0.0124 (0.0086)	0.147
ADHD9_often_motor9	-0.0166 (0.0123)	0.177	-0.0162 (0.0123)	0.186
ADHD9_difficulty_with_attention9	0.0099 (0.0093)	0.285	0.0099 (0.0093)	0.287
ADHD9_often_run9	-0.0083 (0.0090)	0.354	-0.0083 (0.0090)	0.357
ADHD9_often_move9	-0.0080 (0.0102)	0.430	-0.0081 (0.0102)	0.428
ADHD9_not_listening9	0.0064 (0.0083)	0.439	0.0066 (0.0083)	0.429
ADHD9_avoid_mental_effort_tasks9	-0.0062 (0.0084)	0.463	-0.0060 (0.0084)	0.475
ADHD9_difficulty_calm9	-0.0096 (0.0147)	0.511	-0.0092 (0.0147)	0.528
ADHD9_difficulty_organizing_tasks9	-0.0074 (0.0119)	0.534	-0.0077 (0.0119)	0.521
ADHD9_difficulty_following_instructions9	0.0069 (0.0120)	0.563	0.0068 (0.0120)	0.571
ADHD9_easily_bored9	-0.0038 (0.0070)	0.583	-0.0040 (0.0070)	0.565
ADHD9_easily_distracted9	-0.0039 (0.0076)	0.604	-0.0039 (0.0076)	0.604
ADHD9_difficulty_hold_still9	0.0043 (0.0084)	0.609	0.0044 (0.0084)	0.605
ADHD9_talk_excess9	0.0031 (0.0061)	0.611	0.0028 (0.0061)	0.642
ADHD9_often_lose_things9	-0.0014 (0.0073)	0.847	-0.0014 (0.0073)	0.846
ADHD9_difficulty_waiting9	-0.0001 (0.0081)	0.993	0.0001 (0.0081)	0.994
Autism-related traits, age 9				
ASD9_difficulty_converse9	-0.0491 (0.0212)	0.0208	-0.0485 (0.0212)	0.0222
ASD9_dislike_change9	0.0167 (0.0085)	0.0492	0.0166 (0.0085)	0.0504
ASD9_difficulty_socialize9	0.0324 (0.0169)	0.0554	0.0316 (0.0169)	0.062
ASD9_absorbed_details9	0.0163 (0.0102)	0.109	0.0161 (0.0102)	0.112
ASD9_difficulty_pretend_play9	-0.0201 (0.0164)	0.221	-0.0205 (0.0164)	0.213
ASD9_talk_too_high_low9	0.0107 (0.0087)	0.222	0.0111 (0.0087)	0.203
ASD9_repeat_words9	-0.0154 (0.0126)	0.222	-0.0153 (0.0126)	0.227
ASD9_absorbed_problems9	-0.0192 (0.0167)	0.252	-0.0191 (0.0167)	0.254
ASD9_easily_influenced9	0.0060 (0.0064)	0.350	0.0059 (0.0064)	0.362
ASD9_difficulty_on_track9	0.0099 (0.0106)	0.351	0.0101 (0.0106)	0.339
ASD9_absorbed_own9	0.0087 (0.0101)	0.391	0.0088 (0.0101)	0.385
ASD9_delay_language9	0.0063 (0.0074)	0.391	0.0061 (0.0074)	0.405
ASD9_own_terms9	0.0084 (0.0122)	0.492	0.0082 (0.0122)	0.503

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
ASD9_difficulty_express9	-0.0160 (0.0242)	0.510	-0.0156 (0.0242)	0.520
ASD9_difficulty_expect_behavior9	0.0072 (0.0165)	0.661	0.0075 (0.0165)	0.647
ASD9_strange_movements9	-0.0020 (0.0118)	0.866	-0.0017 (0.0118)	0.886
ASD9_uninterested_sharing9	-0.0013 (0.0222)	0.952	-0.0014 (0.0222)	0.950
Adverse experiences, age 15				
ACE15_emotional_bullying15	0.0085 (0.0061)	0.164	0.0084 (0.0061)	0.167
ACE15_bullying_time15	0.0055 (0.0040)	0.166	0.0054 (0.0040)	0.172
ACE15_bullying_by_num15	-0.0051 (0.0058)	0.380	-0.0049 (0.0058)	0.390
ACE15_rumours_bullying15	0.0061 (0.0073)	0.406	0.0062 (0.0073)	0.394
ACE15_tease_bullying15	0.0047 (0.0068)	0.493	0.0047 (0.0068)	0.486
ACE15_bullied_often15	-0.0040 (0.0070)	0.569	-0.0038 (0.0070)	0.593
ACE15_other_bullying15	-0.0031 (0.0064)	0.635	-0.0033 (0.0064)	0.613
Adverse experiences, age 18				
ACE18_emotional_abuse18	-0.0752 (0.0066)	< 0.001	-0.0754 (0.0066)	< 0.001
ACE18_hate_crime18	-0.0544 (0.0175)	0.00197	-0.0542 (0.0175)	0.00203
ACE18_other_abuse18	0.2684 (0.3643)	0.461	0.2695 (0.3641)	0.459
ACE18_witness_crime18	-0.0020 (0.0103)	0.847	-0.0017 (0.0103)	0.871
Substance use, age 18				
SUD18_cigarettes18	0.0107 (0.0021)	< 0.001	0.0107 (0.0021)	< 0.001
SUD18_drugs_often18	0.0216 (0.0083)	0.00912	0.0216 (0.0083)	0.00907
SUD18_alcohol_often18	-0.0038 (0.0041)	0.359	-0.0037 (0.0041)	0.372
SUD18_snuff18	0.0008 (0.0017)	0.660	0.0007 (0.0017)	0.667
Parental socioeconomic indicators				
SES_birth_country_father	-0.0187 (0.0071)	0.00832	-0.0186 (0.0071)	0.00837
SES_education_mother	0.0035 (0.0034)	0.306	0.0036 (0.0034)	0.287
SES_birth_country_mother	0.0072 (0.0080)	0.370	0.0073 (0.0080)	0.358
SES_education_father	0.0012 (0.0029)	0.674	0.0011 (0.0029)	0.692
Sex				
SEX	0.0417 (0.0068)	< 0.001	0.0416 (0.0068)	< 0.001
Genotyping batch × principal component				
batch_PSYC_x_PC2	-0.3105 (0.2287)	0.175	-0.2955 (0.2286)	0.196
batch_GSA3_x_PC1	-0.4600 (0.3846)	0.232	-0.4395 (0.3848)	0.254
batch_PSYC_x_PC1	0.2577 (0.2392)	0.281	0.2593 (0.2390)	0.278
batch_GSA3_x_PC2	0.1085 (0.6516)	0.868	0.0929 (0.6513)	0.887
Interactions with SUD15				
SUD15 score × ACE15 score	-0.0039 (0.0030)	0.192	-0.0039 (0.0030)	0.199
SUD15 score × ACE18 score	-0.0025 (0.0028)	0.373	-0.0025 (0.0028)	0.369
SUD15 score × ADHD9 score	0.0013 (0.0028)	0.647	0.0013 (0.0028)	0.645
SUD15 score × ASD9 score	0.0046 (0.0032)	0.146	0.0046 (0.0032)	0.143
SUD15 score × SCZ15 score	-0.0064 (0.0027)	0.0169	-0.0063 (0.0027)	0.0182
SUD15 score × SES score	0.0016 (0.0032)	0.625	0.0016 (0.0032)	0.611
SUD15 score × SEX	0.0043 (0.0068)	0.527	0.0041 (0.0068)	0.546
SUD15 score × SUD18 score	0.0014 (0.0029)	0.626	0.0014 (0.0029)	0.623
SUD15 score × PRS_Amygdala_excitatory (z)	0.0026 (0.0046)	0.573	0.0026 (0.0046)	0.577

Term	GLMM β (SE)	GLMM p	GAMM β (SE)	GAMM p
SUD15 score × PRS_CGE_interneuron (z)	0.0012 (0.0042)	0.781	0.0013 (0.0042)	0.761
SUD15 score × PRS_Cerebellar_inhibitory (z)	-0.0055 (0.0037)	0.140	-0.0056 (0.0037)	0.132
SUD15 score × PRS_Deep_layer_corticothalamic (z)	0.0071 (0.0041)	0.082	0.0072 (0.0041)	0.0746
SUD15 score × PRS_Deep_layer_intratelencepha (z)	0.0056 (0.0048)	0.243	0.0057 (0.0048)	0.240
SUD15 score × PRS_Eccentric_medium_spiny_neu (z)	-0.0016 (0.0043)	0.708	-0.0016 (0.0043)	0.708
SUD15 score × PRS_Hippocampal_CA1_3 (z)	0.0068 (0.0048)	0.161	0.0065 (0.0048)	0.183
SUD15 score × PRS_Hippocampal_CA4 (z)	-0.0081 (0.0043)	0.058	-0.0078 (0.0043)	0.0663
SUD15 score × PRS_Hippocampal_dentate_gyrus (z)	0.0010 (0.0038)	0.804	0.0008 (0.0038)	0.831
SUD15 score × PRS_LAMP5_LHX6_and_Chandelier (z)	0.0035 (0.0048)	0.468	0.0035 (0.0048)	0.474
SUD15 score × PRS_MGE_interneuron (z)	-0.0016 (0.0043)	0.716	-0.0017 (0.0043)	0.692
SUD15 score × PRS_Medium_spiny_neuron (z)	-0.0036 (0.0044)	0.420	-0.0035 (0.0044)	0.424
SUD15 score × PRS_Miscellaneous (z)	-0.0002 (0.0043)	0.960	-0.0003 (0.0043)	0.950
SUD15 score × PRS_Thalamic_excitatory (z)	0.0021 (0.0032)	0.515	0.0021 (0.0032)	0.513
SUD15 score × PRS_Upper_layer_intratelenceph (z)	-0.0004 (0.0042)	0.931	-0.0002 (0.0042)	0.953
SUD15 score × PRS_all (z)	0.0018 (0.0038)	0.631	0.0017 (0.0038)	0.649

Table S3c. PRS smooth terms of the additive mixed models

Smooth	PLE-HM edf	PLE-HM F	PLE-HM p	DEP edf	DEP F	DEP p
s(PRS_Amygdala_excitatory)	1.66	1.75	0.186	1.00	0.25	0.618
s(PRS_Cerebellar_inhibitory)	2.44	0.83	0.450	1.72	1.03	0.384
s(PRS_CGE_interneuron)	1.87	1.25	0.323	1.92	0.74	0.608
s(PRS_Deep_layer_corticothalamic)	1.00	0.20	0.655	1.00	0.53	0.468
s(PRS_Deep_layer_intratelencepha)	1.00	1.27	0.261	1.00	0.62	0.433

Smooth	PLE-HM edf	PLE-HM F	PLE-HM p	DEP edf	DEP F	DEP p
s(PRS_Eccentric_medium_spiny_neu)	2.47	1.03	0.337	1.13	0.99	0.385
s(PRS_Hippocampal_CA1_3)	3.54	2.56	0.0497	1.00	0.00	0.980
s(PRS_Hippocampal_CA4)	1.00	0.30	0.582	1.00	1.59	0.207
s(PRS_Hippocampal_dentate_gyrus)	1.00	0.02	0.899	1.64	0.49	0.606
s(PRS_LAMP5_LHX6_and_Chandelier)	1.28	0.14	0.890	1.00	0.14	0.707
s(PRS_Medium_spiny_neuron)	1.00	0.07	0.796	1.00	0.80	0.371
s(PRS_MGE_interneuron)	1.00	0.38	0.536	1.00	0.37	0.542
s(PRS_Miscellaneous)	1.00	3.87	0.0493	1.00	0.07	0.794
s(PRS_Thalamic_excitatory)	1.00	0.15	0.698	1.32	0.16	0.865
s(PRS_Upper_layer_intratelenceph)	1.00	0.23	0.632	1.00	0.01	0.927
s(PRS_all)	1.00	13.10	< 0.001	1.00	6.28	0.0122
Twin-pair random intercept (s(cmpair), random-effect basis)	353.96	0.22	< 0.001	460.89	0.31	< 0.001

edf, effective degrees of freedom (1 = linear).

Table S4. Predictors by group

Variable names as in the analysis data. Items combined during screening (see Methods): SUD15_OtherDrugs15 (11 illicit drugs), SUD15_Alcohol15 (beer and other alcohol), ACE15_other_bullying15 (six types of bullying) and ACE18_other_abuse18 (five types of abuse or violence). SCZ15_PROD_seen_hallucinations9 was measured at age 15; the suffix 9 in its name is a labelling error in the analysis data.

Group	n	Variables
Substance use, age 15 (anchor) (SUD15)	6	SUD15_Alcohol15, SUD15_Cannabis15, SUD15_Cigarettes15, SUD15_OtherDrugs15, SUD15_Painkillers_opioids15, SUD15_Snuff15
Polygenic risk scores (PRS)	16	PRS_Amygdala_excitatory, PRS_CGE_interneuron, PRS_Cerebellar_inhibitory, PRS_Deep_layer_corticothalamic, PRS_Deep_layer_intratelencepha, PRS_Eccentric_medium_spiny_neu, PRS_Hippocampal_CA1_3, PRS_Hippocampal_CA4, PRS_Hippocampal_dentate_gyrus, PRS_LAMP5_LHX6_and_Chandelier, PRS_MGE_interneuron, PRS_Medium_spiny_neuron, PRS_Miscellaneous, PRS_Thalamic_excitatory, PRS_Upper_layer_intratelenceph, PRS_all

Group	n	Variables
Symptoms, age 15 (SCZ15)	23	SCZ15_Extreme_excitement15, SCZ15_Hear_voices15, SCZ15_Irritable15, SCZ15_Not_tired15, SCZ15_Others_Read_thoughts15, SCZ15_PROD_seen_hallucinations9, SCZ15_Racing_thoughts15, SCZ15_Rage_attacks15, SCZ15_Read_others_minds15, SCZ15_Seen_hallucinations15, SCZ15_Sexual_inappropriate15, SCZ15_Special_messages15, SCZ15_Special_powers15, SCZ15_Spied15, SCZ15_Talking_fast15, SCZ15_Too_much_energy15, SCZ15_Under_control_special_power15, SCZ15_Unrealistic_abilities15, SCZ15_easily_scared15, SCZ15_headaches15, SCZ15_lose_confidence15, SCZ15_unhappy15, SCZ15_worry15
ADHD symptoms, age 9 (ADHD9)	19	ADHD9_avoid_mental_effort_tasks9, ADHD9_difficulty_calm9, ADHD9_difficulty_following_instructions9, ADHD9_difficulty_hold_still9, ADHD9_difficulty_organizing_tasks9, ADHD9_difficulty_waiting9, ADHD9_difficulty_with_attention9, ADHD9_easily_bored9, ADHD9_easily_distracted9, ADHD9_fail_close_attention9, ADHD9_not_listening9, ADHD9_often_blurt_out9, ADHD9_often_forgetful9, ADHD9_often_interrupt9, ADHD9_often_lose_things9, ADHD9_often_motor9, ADHD9_often_move9, ADHD9_often_run9, ADHD9_talk_excess9
Autism-related traits, age 9 (ASD9)	17	ASD9_absorbed_details9, ASD9_absorbed_own9, ASD9_absorbed_problems9, ASD9_delay_language9, ASD9_difficulty_converse9, ASD9_difficulty_expect_behavior9, ASD9_difficulty_express9, ASD9_difficulty_on_track9, ASD9_difficulty_pretend_play9, ASD9_difficulty_socialize9, ASD9_dislike_change9, ASD9_easily_influenced9, ASD9_own_terms9, ASD9_repeat_words9, ASD9_strange_movements9, ASD9_talk_too_high_low9, ASD9_uninterested_sharing9
Adverse experiences, age 15 (ACE15)	7	ACE15_bullied_often15, ACE15_bullying_by_num15, ACE15_bullying_time15, ACE15_emotional_bullying15, ACE15_other_bullying15, ACE15_rumours_bullying15, ACE15_tease_bullying15

Group	n	Variables
Adverse experiences, age 18 (ACE18)	4	`ACE18_emotional_abuse18`, `ACE18_hate_crime18`, `ACE18_other_abuse18`, `ACE18_witness_crime18`
Substance use, age 18 (SUD18)	4	`SUD18_alcohol_often18`, `SUD18_cigarettes18`, `SUD18_drugs_often18`, `SUD18_snuff18`
Parental socioeconomic indicators (SES)	4	`SES_birth_country_father`, `SES_birth_country_mother`, `SES_education_father`, `SES_education_mother`
Sex (SEX)	1	`SEX`
Genotyping batch × principal component (batch)	4	`batch_GSA3_x_PC1`, `batch_GSA3_x_PC2`, `batch_PSYC_x_PC1`, `batch_PSYC_x_PC2`
Total	**105**	

Table S5. Selection of the fusion depths

After the hyperparameter search (Table S1), the depth of each fusion block was varied from 1 to 5 while the other blocks were kept at the uniform depth from the search (3 for PLE-HM, 2 for DEP). All runs used the seed-42 training set, with the same 80/20 training/validation split as the search. The depth with the lowest validation RMSE was taken for each block. The resulting per-block configuration and the uniform configuration were then compared on the seed-42 test set. Per-block depths were adopted for PLE-HM, where they had the better test performance, and the uniform depth was kept for DEP. This choice used test-set results (see Methods). Bold: lowest validation RMSE per block.

PLE-HM (other blocks at the uniform depth 3; validation RMSE × 100)

Fusion block	Depth 1	Depth 2	Depth 3	Depth 4	Depth 5	Selected
PRS	9.030	9.081	8.993	8.993	**8.887**	5
SCZ15	8.930	**8.875**	8.993	8.906	8.971	2
ADHD9	**8.899**	9.140	8.993	8.995	8.942	1
ASD9	8.956	8.911	8.993	9.048	**8.888**	5
ACE15	9.014	9.066	8.993	**8.964**	9.005	4
ACE18	**8.766**	8.891	8.993	8.975	9.109	1
SUD18	**8.767**	8.890	8.993	8.976	9.119	1
SES	**8.766**	8.893	8.993	8.982	9.115	1
SEX	8.953	8.929	8.993	9.050	**8.870**	5

Configuration	Depths	Validation RMSE	Validation R^2	Test RMSE (seed 42)	Test R^2 (seed 42)
Per-block	[5, 2, 1, 5, 4, 1, 1, 1, 5]	0.0898	0.204	0.0863	0.262
Uniform	3	0.0899	0.201	0.0891	0.213

DEP (other blocks at the uniform depth 2; validation RMSE × 100)

Fusion block	Depth 1	Depth 2	Depth 3	Depth 4	Depth 5	Selected
PRS	**15.004**	15.044	15.223	15.354	15.042	1
SCZ15	15.312	15.044	15.333	**14.961**	15.308	4
ADHD9	15.478	**15.044**	15.339	15.358	15.070	2

Fusion block	Depth 1	Depth 2	Depth 3	Depth 4	Depth 5	Selected
ASD9	**15.024**	15.044	15.042	15.436	15.186	1
ACE15	**14.984**	15.044	15.450	15.098	15.012	1
ACE18	15.327	**15.044**	15.191	15.152	15.056	2
SUD18	15.318	**15.044**	15.186	15.157	15.052	2
SES	15.322	**15.044**	15.189	15.160	15.048	2
SEX	15.036	15.044	**14.942**	15.512	15.186	3

Configuration	Depths	Validation RMSE	Validation R^2	Test RMSE (seed 42)	Test R^2 (seed 42)
Per-block	[1, 4, 2, 1, 1, 2, 2, 2, 3]	0.1536	0.213	0.1551	0.201
Uniform	2	0.1504	0.245	0.1531	0.221

Table S6. Test-set performance on all metrics

Mean ± SD over the five test sets, with all predictors. R^2, coefficient of determination (1 − SSE/SST); RMSE, root-mean-squared error; MAE, mean absolute error; Spearman ρ, rank correlation between predicted and observed scores. R^2 and RMSE are also given in Table 3 of the main text.

Model	R^2	RMSE	MAE	Spearman ρ
PLE-HM				
DCMFNet	0.2058 ± 0.0189	0.0903 ± 0.0013	0.0712 ± 0.0004	0.4497 ± 0.0193
GLMM	0.2279 ± 0.0156	0.0890 ± 0.0014	0.0700 ± 0.0012	0.4641 ± 0.0183
GAMM	0.2274 ± 0.0153	0.0891 ± 0.0014	0.0699 ± 0.0011	0.4633 ± 0.0180
Linear regression	0.2393 ± 0.0212	0.0884 ± 0.0017	0.0695 ± 0.0012	0.4728 ± 0.0215
Elastic net	0.2603 ± 0.0169	0.0871 ± 0.0016	0.0691 ± 0.0010	0.4915 ± 0.0182
Gradient boosting	0.2441 ± 0.0188	0.0881 ± 0.0016	0.0700 ± 0.0008	0.4792 ± 0.0194
DEP				
DCMFNet	0.1984 ± 0.0208	0.1562 ± 0.0017	0.1242 ± 0.0019	0.4509 ± 0.0135
GLMM	0.2025 ± 0.0220	0.1559 ± 0.0024	0.1226 ± 0.0016	0.4460 ± 0.0154
GAMM	0.1998 ± 0.0230	0.1561 ± 0.0026	0.1228 ± 0.0017	0.4421 ± 0.0163
Linear regression	0.2150 ± 0.0253	0.1546 ± 0.0026	0.1216 ± 0.0017	0.4531 ± 0.0203
Elastic net	0.2373 ± 0.0146	0.1524 ± 0.0024	0.1197 ± 0.0012	0.4733 ± 0.0193
Gradient boosting	0.2287 ± 0.0163	0.1533 ± 0.0027	0.1203 ± 0.0017	0.4645 ± 0.0194

Table S7. Five most important predictors in DCMFNet and the GLMM

DCMFNet: mean absolute SHAP value over the five test sets (SD across splits), and its share of the summed mean absolute SHAP values of all 105 predictors. GLMM: predictors ranked by the absolute t value of their main effect in the model fitted on the first split (seed 42; interaction terms not ranked); the full coefficients are in Table S3. Per-predictor SHAP values were not computed for the GLMM. Both columns describe what the fitted models use for prediction, not causal effects. Variable names as in Table S4.

Rank	DCMFNet predictor	Mean |SHAP| (SD)	Share	GLMM predictor	t
PLE-HM					
1	`ACE18_emotional_abuse18`	0.0119 (0.0044)	7.2%	`ACE18_emotional_abuse18`	−9.13

Rank	DCMFNet predictor	Mean |SHAP| (SD)	Share	GLMM predictor	t
2	SUD18_cigarettes18	0.0082 (0.0021)	5.0%	SCZ15_Too_much_energy15	4.49
3	SCZ15_Talking_fast15	0.0072 (0.0010)	4.4%	SUD18_alcohol_often18	3.92
4	SCZ15_Too_much_energy15	0.0070 (0.0012)	4.3%	SCZ15_Seen_hallucinations15	4.87
5	SUD18_alcohol_often18	0.0061 (0.0008)	3.7%	SCZ15_Talking_fast15	3.64
DEP					
1	ACE18_emotional_abuse18	0.0296 (0.0036)	11.0%	ACE18_emotional_abuse18	−11.37
2	SCZ15_unhappy15	0.0174 (0.0018)	6.5%	SCZ15_unhappy15	6.86
3	SCZ15_headaches15	0.0170 (0.0008)	6.4%	SCZ15_headaches15	6.21
4	SCZ15_worry15	0.0140 (0.0020)	5.2%	SEX	6.09
5	SEX	0.0133 (0.0020)	5.0%	SUD18_cigarettes18	5.19